

Planning multi-functional forest management at the landscape scale

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Introduction

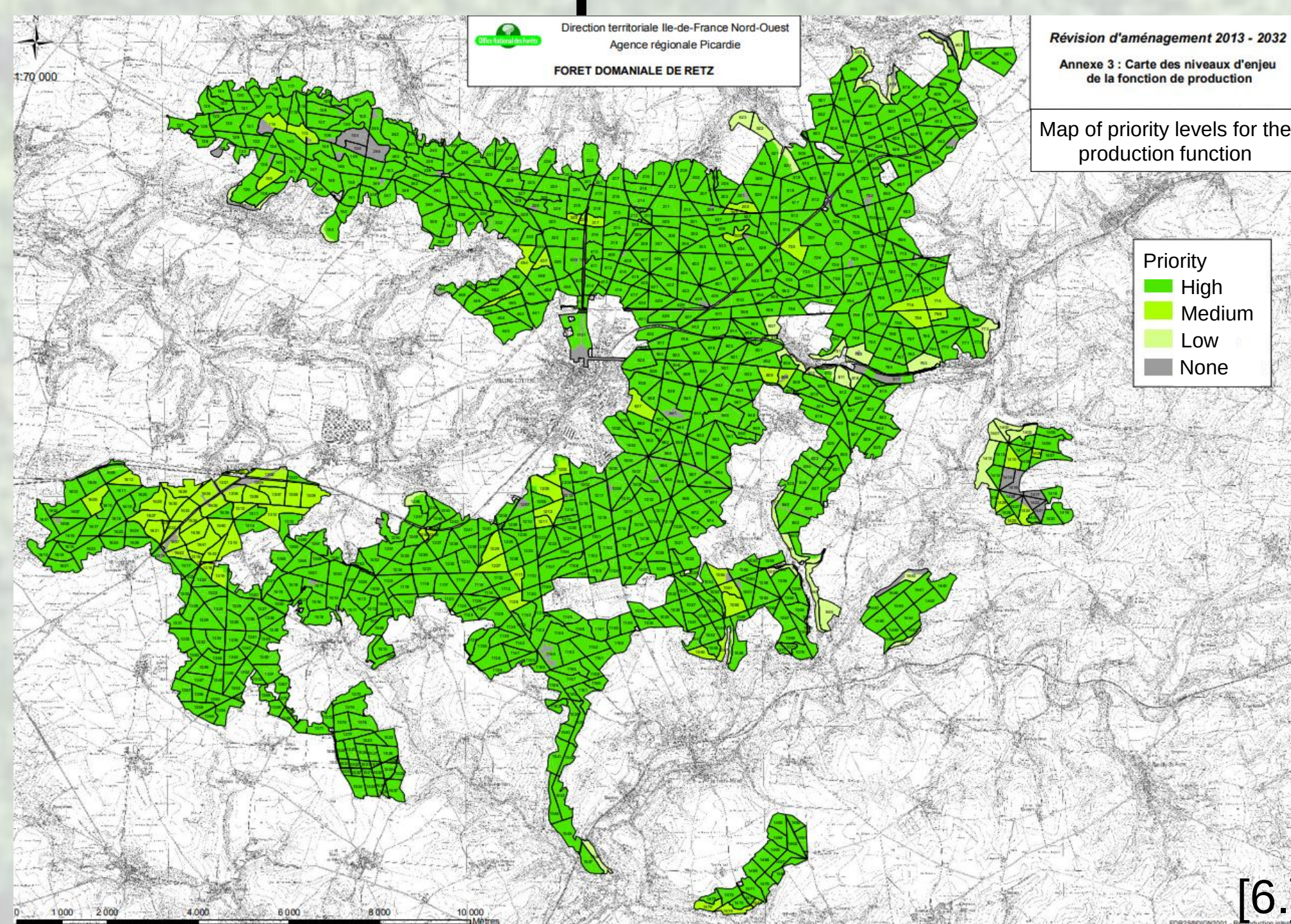
Managing multi-functional forests—for timber, carbon storage, and biodiversity—requires planning at both the stand and landscape scales. While stand-level management can enhance local biodiversity, it may lead to homogenization across the landscape, reducing habitat diversity and increasing vulnerability to disturbances [1]. Landscape-scale approaches are better suited to address these broader ecological dynamics and support more resilient forest systems [2].

However, implementing effective landscape-scale planning is complex. It involves not only deciding what management actions to take but also where to apply them across heterogeneous forested areas.

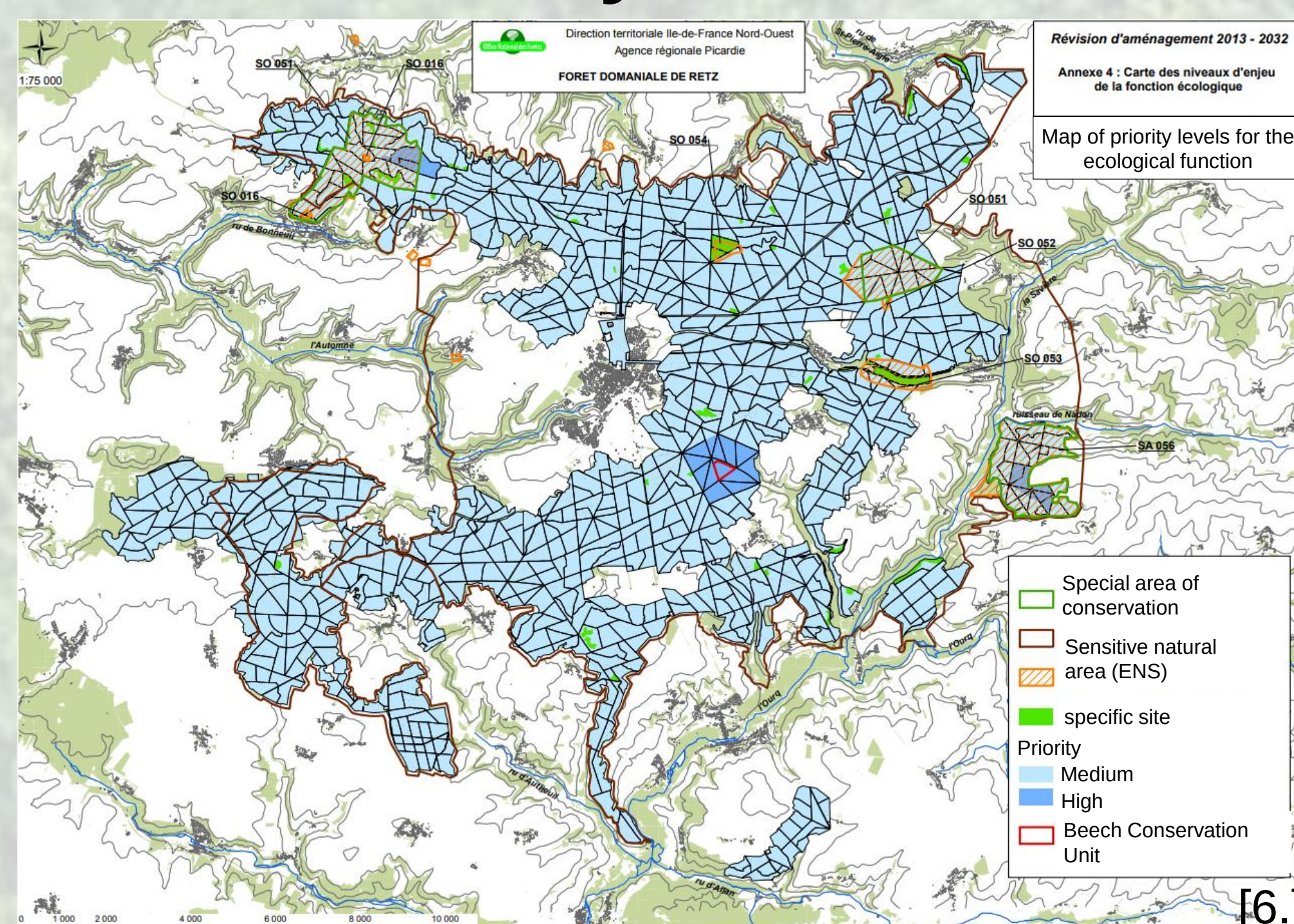
To address these challenges, innovative tools from complex systems dynamics and control theory are needed. These methods can help explore a broad set of planning strategies that account for spatial structure, trade-offs, and the need for multi-functionality in forest landscapes..

Spatial functions

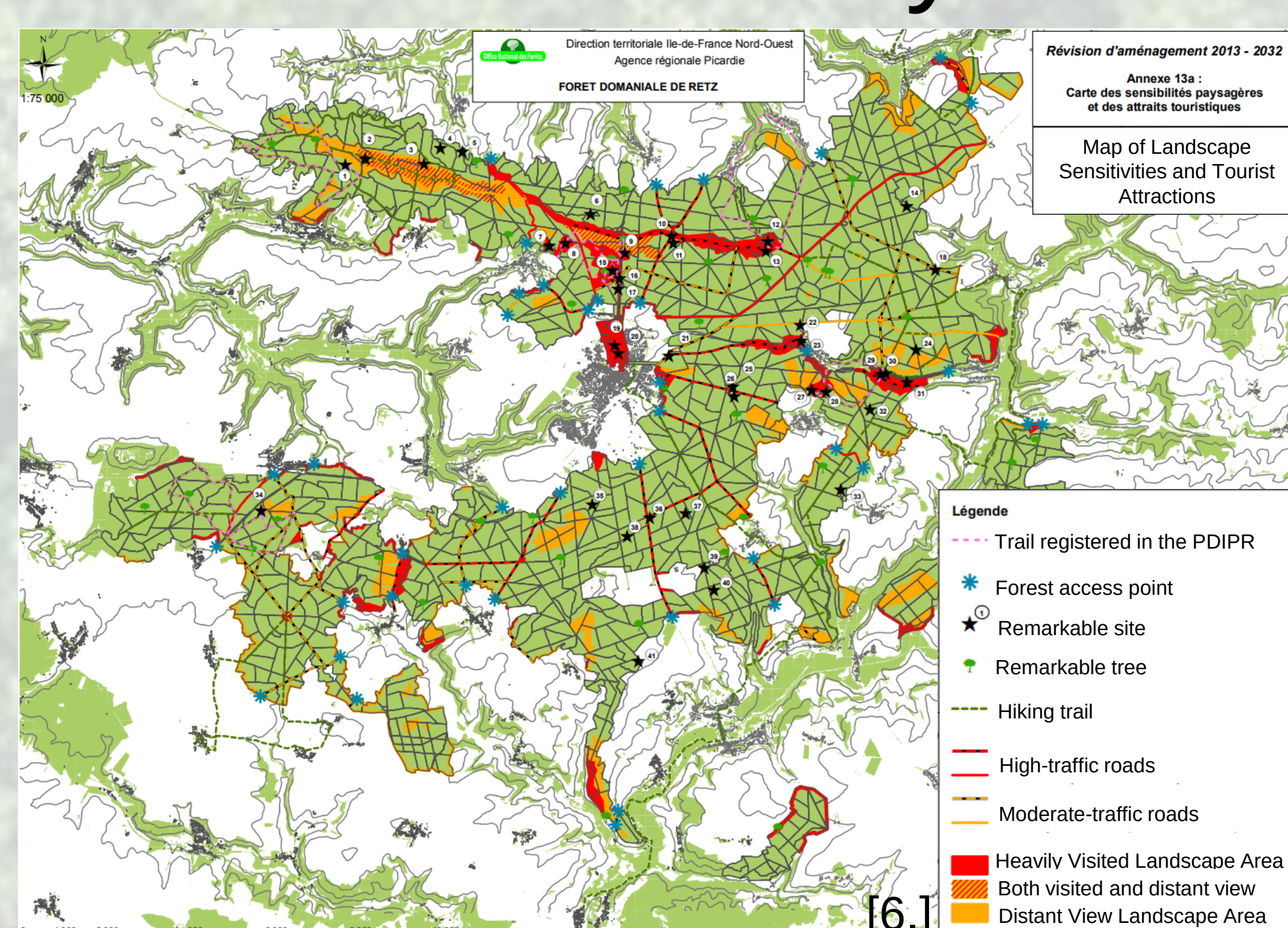
Wood production



Biodiversity and habitat



Social utility



The maps reveal the spatial complexity of forest management challenges.

- Diverse functions and goals to address
- Uneven spatial scales: some very local, others global
- Overlapping zones, potential hotspots for conflict

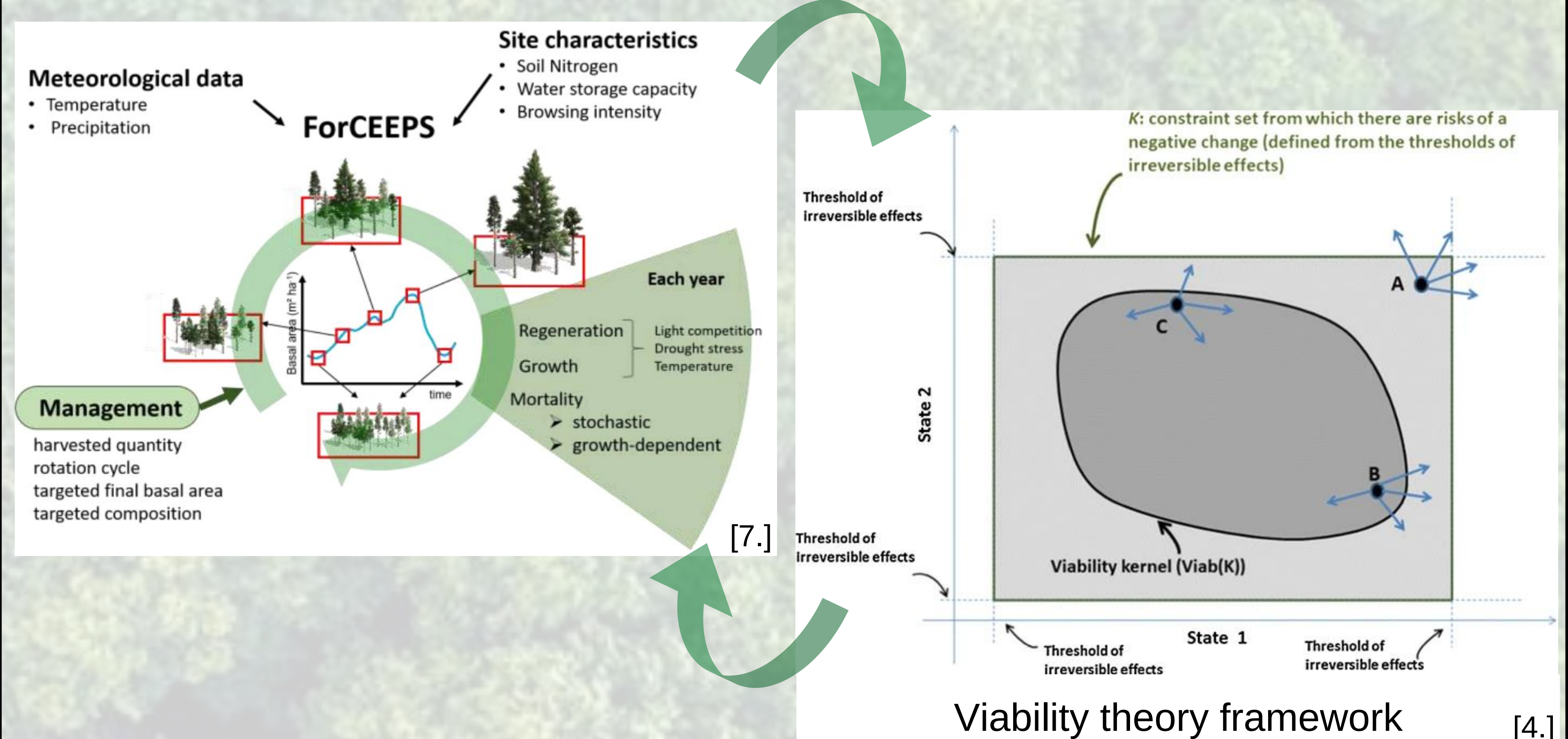
Method development

Challenges

The complexity of terrain and ecosystems across large landscapes presents significant challenges for management methods, including:

- Modeling ecosystems accurately at broad spatial scales
- Managing the computational complexity due to a large number of possible management actions
- Meeting multiple, often conflicting constraints simultaneously
- Adaptability of tools for different use (limit of optimality)

Goals



- Extend the mechanistic forest model ForCEEPS to predict dynamics at the forest scale
- Apply viability theory to formulate forest management as a control problem under multiple constraints

Conclusion

Managing forests at the landscape scale is challenging due to the complexity of ecosystems, the diversity of services they provide, and the need to balance production, biodiversity, social demands, and climate adaptation.

New tools like control theory can help by revealing system dynamics and offering flexible pathways, allowing managers to adapt strategies to their local context [4].

To make this possible, new governance structures will be needed—ones that support collaboration and long-term planning across scales [5].

Bibliography

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